Week-by-Week Pregnancy Tracker Implementation Prompt

Context & Project Standards

You are implementing a week-by-week pregnancy tracker for SageNest, a privacy-first pregnancy calculator web app. Before starting ANY code, you MUST read these project documentation files in this order:

1. **SECURITY.md** - Review sections 1.1-1.4 (Anti-Mess Protocol, Pattern Matching, Dependency Minimalism, Complexity Check)
2. **frontend.md** - Review the complete design system, color palette, typography scale, component library, and spacing system
3. **codex.md** - Review behavioral guidelines for LLM coding
4. **newtool.md** - Review the tool addition guide (though this is more content than tool)
5. **README.md** - Review the tech stack and build commands

Feature Overview

Implement a comprehensive week-by-week pregnancy guide covering weeks 1-40, with:
- A hub page at `/pregnancy-week-by-week` with visual week selector
- 40 individual week pages at `/pregnancy-week-by-week/week-{1-40}`
- Integration with existing due date calculator
- Full SEO optimization with JSON-LD structured data
- Accessibility compliance (WCAG 2.1 AA)

Implementation Plan (4 Phases)

Phase 1: Data Structure & Core Infrastructure

File: `src/data/pregnancyWeeks.ts`

Create a TypeScript data structure following these patterns from `src/data/blogPosts.ts` and `src/data/tools.ts`:

```typescript
export interface PregnancyWeek {
  weekNumber: number;
  trimester: 1 | 2 | 3;
  title: string; // e.g., "Week 12: Fetal Development Milestone"
  babyDevelopment: {
    summary: string; // 2-3 sentences
    size: {
      comparison: string; // e.g., "lime"
      lengthCm: number;
```

```
      weightGrams: number;
    };
    keyMilestones: string[]; // 3-5 bullet points
  };
  maternalChanges: {
    symptoms: string[]; // 3-5 common symptoms
    bodyChanges: string; // 1-2 sentences
  };
  whatToDo: {
    appointments: string[]; // Upcoming appointments if any
    preparation: string[]; // 2-3 actionable items
  };
  medicalGuidance: {
    whenToCallDoctor: string[]; // 3-4 red flags
    screeningTests?: string[]; // If applicable this week
  };
  funFact: string; // Engaging, short fact
}

export const pregnancyWeeks: PregnancyWeek[] = [
  {
    weekNumber: 1,
    trimester: 1,
    title: "Week 1: Conception Preparation",
    babyDevelopment: {
      summary: "Week 1 is technically before conception. Your body is preparing for ovulation,
and pregnancy dating begins from the first day of your last menstrual period.",
      size: {
        comparison: "Not yet conceived",
        lengthCm: 0,
        weightGrams: 0
      },
      keyMilestones: [
        "Menstrual period begins",
        "Uterine lining sheds in preparation for new cycle",
        "Follicle development begins in ovaries"
      ]
    },
    maternalChanges: {
      symptoms: [
        "Menstrual bleeding",
        "Mild cramping",
        "Lower energy levels"
      ],
```

```
      bodyChanges: "Your body is in its menstrual phase. Hormone levels are at their lowest, preparing to rise for ovulation."
    },
    whatToDo: {
      appointments: [],
      preparation: [
        "Start taking prenatal vitamins with folic acid (400-800 mcg)",
        "Avoid alcohol and smoking if trying to conceive",
        "Track your cycle to identify ovulation"
      ]
    },
    medicalGuidance: {
      whenToCallDoctor: [
        "Extremely heavy bleeding (soaking through a pad every hour)",
        "Severe pain that doesn't respond to over-the-counter medication",
        "Fever above 100.4°F (38°C)"
      ]
    },
    funFact: "Pregnancy is dated from your last menstrual period, even though conception hasn't occurred yet. This means you're 'pregnant' for about two weeks before actual conception!"
  },
  // ... continue for weeks 2-40
];

// Helper functions
export const getWeekByNumber = (weekNumber: number): PregnancyWeek | undefined => {
  return pregnancyWeeks.find(w => w.weekNumber === weekNumber);
};

export const getWeeksByTrimester = (trimester: 1 | 2 | 3): PregnancyWeek[] => {
  return pregnancyWeeks.filter(w => w.trimester === trimester);
};

export const getTrimesterForWeek = (weekNumber: number): 1 | 2 | 3 => {
  if (weekNumber <= 13) return 1;
  if (weekNumber <= 27) return 2;
  return 3;
};
```

Content Requirements for All 40 Weeks:
- Week 1-13: Focus on early development, morning sickness, first trimester screening
- Week 14-27: Emphasis on fetal movement, anatomy scan, glucose testing
- Week 28-40: Birth preparation, weekly appointments, labor signs

Specific Milestone Weeks to Emphasize:
- Week 12: NT scan, end of first trimester
- Week 20: Anatomy scan
- Week 24: Viability milestone
- Week 28: Third trimester begins
- Week 36: Group B strep test, weekly appointments
- Week 37: Full term
- Week 40: Due date

Phase 2: Hub Page Component

File: `src/pages/PregnancyWeekByWeek.tsx`

Create the main hub page following the pattern from `src/pages/SimilarTools.tsx` and `src/pages/Calculator.tsx`:

```typescript
import { useMemo, useState } from 'react';
import { Link } from 'react-router-dom';
import { SEOHead } from '../components/SEOHead';
import { pregnancyWeeks, getTrimesterForWeek } from '../data/pregnancyWeeks';

export const PregnancyWeekByWeekPage = () => {
  const [activeTab, setActiveTab] = useState<1 | 2 | 3 | 'all'>('all');
  const [currentWeekInput, setCurrentWeekInput] = useState('');

  const filteredWeeks = useMemo(() => {
    if (activeTab === 'all') return pregnancyWeeks;
    return pregnancyWeeks.filter(w => w.trimester === activeTab);
  }, [activeTab]);

  const jumpToWeek = () => {
    const week = Number(currentWeekInput);
    if (week >= 1 && week <= 40) {
      window.location.href = `/pregnancy-week-by-week/week-${week}`;
    }
  };

  const jsonLd = {
    '@context': 'https://schema.org',
    '@type': 'WebPage',
```

```
    name: 'Week-by-Week Pregnancy Guide',
    description: 'Complete week-by-week pregnancy guide covering fetal development, maternal
changes, and medical guidance for all 40 weeks.',
    url: 'https://sagenest.pages.dev/pregnancy-week-by-week'
  };

  return (
    <main id="main-content" className="container">
      <SEOHead
        title="Week-by-Week Pregnancy Guide"
        description="Track your pregnancy week by week with detailed fetal development,
symptoms, medical guidance, and preparation tips for all 40 weeks."
        canonicalPath="/pregnancy-week-by-week"
        jsonLd={jsonLd}
      />

      <section className="hero-section">
        <div className="tool-header tool-header-inline">
          <div className="trust-badge">Comprehensive pregnancy reference</div>
        </div>
        <h1 className="hero-title">Week-by-Week <span>Pregnancy Guide</span></h1>
        <p className="hero-description">
          Follow your pregnancy journey with detailed weekly guides covering fetal development,
maternal changes, and medical milestones.
        </p>
      </section>

      {/* Quick Week Finder */}
      <div className="calculator-card" style={{ marginBottom: 'var(--space-xl)' }}>
        <div className="calculator-header">
          <h2 className="calculator-title">Find Your Week</h2>
          <p className="calculator-subtitle">Enter your current pregnancy week to jump directly to
your guide</p>
        </div>

        <div style={{ display: 'flex', gap: 'var(--space-md)', alignItems: 'flex-end', maxWidth: '400px',
margin: '0 auto' }}>
          <div className="field-row" style={{ flex: 1, marginBottom: 0 }}>
            <label htmlFor="week-input">Current Week (1-40)</label>
            <input
              id="week-input"
              type="number"
              min={1}
              max={40}
```

```
            value={currentWeekInput}
            onChange={(e) => setCurrentWeekInput(e.target.value)}
            placeholder="e.g., 20"
            onKeyDown={(e) => e.key === 'Enter' && jumpToWeek()}
          />
        </div>
        <button
          className="primary"
          onClick={jumpToWeek}
          disabled={!currentWeekInput || Number(currentWeekInput) < 1 ||
Number(currentWeekInput) > 40}
          style={{ marginBottom: 0 }}
        >
          Go to Week
        </button>
      </div>
    </div>

    {/* Trimester Tabs */}
    <div style={{ display: 'flex', justifyContent: 'center', gap: 'var(--space-sm)', marginBottom:
'var(--space-xl)', flexWrap: 'wrap' }}>
      {[
        { value: 'all' as const, label: 'All Weeks' },
        { value: 1 as const, label: 'First Trimester (1-13)' },
        { value: 2 as const, label: 'Second Trimester (14-27)' },
        { value: 3 as const, label: 'Third Trimester (28-40)' }
      ].map(tab => (
        <button
          key={tab.value}
          type="button"
          onClick={() => setActiveTab(tab.value)}
          style={{
            padding: '0.75rem 1.5rem',
            background: activeTab === tab.value ? 'var(--sage-primary)' : 'var(--off-white)',
            color: activeTab === tab.value ? 'var(--off-white)' : 'var(--charcoal)',
            borderColor: activeTab === tab.value ? 'var(--sage-primary)' : 'var(--border-subtle)',
            width: 'auto'
          }}
        >
          {tab.label}
        </button>
      ))}
    </div>
```

```
{/* Week Grid */}
<div
  className="tools-grid"
  style={{
    gridTemplateColumns: 'repeat(auto-fill, minmax(180px, 1fr))',
    gap: 'var(--space-md)'
  }}
>
  {filteredWeeks.map(week => (
    <Link
      key={week.weekNumber}
      to={`/pregnancy-week-by-week/week-${week.weekNumber}`}
      className="tool-card"
      style={{
        textDecoration: 'none',
        display: 'flex',
        flexDirection: 'column',
        alignItems: 'center',
        justifyContent: 'center',
        minHeight: '140px',
        textAlign: 'center',
        padding: 'var(--space-lg)'
      }}
    >
      <div style={{
        fontSize: '2.5rem',
        fontFamily: "'Playfair Display', serif",
        fontWeight: 700,
        color: 'var(--sage-primary)',
        marginBottom: 'var(--space-xs)'
      }}>
        {week.weekNumber}
      </div>
      <div style={{
        fontSize: '0.75rem',
        fontWeight: 600,
        letterSpacing: '0.05em',
        textTransform: 'uppercase',
        color: 'var(--text-tertiary)',
        marginBottom: 'var(--space-xs)'
      }}>
        Week {week.weekNumber}
      </div>
      <div style={{
```

```
            fontSize: '0.875rem',
            color: 'var(--text-secondary)'
          }}>
            {week.babyDevelopment.size.comparison !== 'Not yet conceived'
              ? `Size: ${week.babyDevelopment.size.comparison}`
              : 'Pre-conception'
            }
          </div>
        </Link>
      ))}
    </div>

    {/* Educational Content Section */}
    <section className="content-section" style={{ marginTop: 'var(--space-2xl)' }}>
      <h2 style={{ textAlign: 'center', marginBottom: 'var(--space-lg)' }}>How to Use This
Guide</h2>
      <div className="info-grid">
        <div className="info-card">
          <h3>Week-by-Week Tracking</h3>
          <p>Each week includes detailed fetal development milestones, maternal body changes,
and symptoms you may experience.</p>
        </div>
        <div className="info-card">
          <h3>Medical Guidance</h3>
          <p>Find important screening tests, prenatal appointments, and red flags that require
immediate medical attention.</p>
        </div>
        <div className="info-card">
          <h3>Preparation Tips</h3>
          <p>Actionable advice for each week to help you prepare for appointments, delivery, and
early parenthood.</p>
        </div>
        <div className="info-card">
          <h3>Evidence-Based</h3>
          <p>All content follows ACOG guidelines and current medical research on fetal
development and prenatal care.</p>
        </div>
      </div>
    </section>

    {/* Integration CTA */}
    <section className="cta-section">
      <p className="eyebrow" style={{ color: 'var(--sage-dark)' }}>Planning Tools</p>
      <h2>Calculate Your Due Date</h2>
```

```
        <p>Not sure which week you're in? Use our due date calculator to find your current week and estimated delivery date.</p>
        <div className="cta-buttons">
          <button onClick={() => window.location.href = '/pregnancy-due-date-calculator'}>
            Calculate Due Date
          </button>
        </div>
      </section>

      <p className="disclaimer">
        This guide is for informational purposes only and does not replace professional medical advice. Always consult your healthcare provider about your specific pregnancy.
      </p>
    </main>
  );
};
```

Phase 3: Individual Week Page Component

File: `src/pages/PregnancyWeekDetail.tsx`

Create the individual week page following patterns from `src/components/BlogPost.tsx`:

```typescript
import { useMemo } from 'react';
import { Link, useParams } from 'react-router-dom';
import { SEOHead } from '../components/SEOHead';
import { getWeekByNumber, getTrimesterForWeek } from '../data/pregnancyWeeks';

export const PregnancyWeekDetailPage = () => {
  const { weekNumber } = useParams<{ weekNumber: string }>();
  const week = useMemo(() => getWeekByNumber(Number(weekNumber)), [weekNumber]);

  if (!week || !weekNumber) {
    return (
      <main className="container">
        <h1>Week Not Found</h1>
        <p>The pregnancy week you're looking for doesn't exist.</p>
        <Link to="/pregnancy-week-by-week">← Back to all weeks</Link>
      </main>
    );
```

```
  }

  const prevWeek = week.weekNumber > 1 ? week.weekNumber - 1 : null;
  const nextWeek = week.weekNumber < 40 ? week.weekNumber + 1 : null;

  const jsonLd = {
    '@context': 'https://schema.org',
    '@type': 'Article',
    headline: `${week.title} - Pregnancy Week ${week.weekNumber}`,
    description: week.babyDevelopment.summary,
    datePublished: '2026-02-01',
    dateModified: '2026-02-20',
    mainEntityOfPage:
`https://sagenest.pages.dev/pregnancy-week-by-week/week-${week.weekNumber}`,
    publisher: {
      '@type': 'Organization',
      name: 'SageNest',
      url: 'https://sagenest.pages.dev'
    }
  };

  return (
    <main id="main-content" className="container article">
      <SEOHead
        title={`Week ${week.weekNumber} of Pregnancy: ${week.title.split(':')[1]?.trim() ||
'Pregnancy Guide'}`}
        description={week.babyDevelopment.summary}
        canonicalPath={`/pregnancy-week-by-week/week-${week.weekNumber}`}
        jsonLd={jsonLd}
      />

      {/* Breadcrumb Navigation */}
      <nav aria-label="Breadcrumb" style={{ marginBottom: 'var(--space-lg)' }}>
        <p style={{ fontSize: '0.875rem', color: 'var(--text-tertiary)' }}>
          <Link to="/pregnancy-week-by-week" style={{ color: 'var(--sage-primary)' }}>
            Week-by-Week Guide
          </Link>
          {' '}/{' '}
          <span>Week {week.weekNumber}</span>
        </p>
      </nav>

      {/* Week Header */}
      <header style={{ marginBottom: 'var(--space-xl)' }}>
```

```
        <div style={{
          display: 'inline-flex',
          alignItems: 'center',
          gap: 'var(--space-sm)',
          padding: 'var(--space-xs) var(--space-md)',
          background: 'var(--sage-softest)',
          borderRadius: 'var(--radius-pill)',
          marginBottom: 'var(--space-md)',
          border: '1px solid var(--sage-light)'
        }}>
          <span style={{
            fontSize: '0.75rem',
            fontWeight: 600,
            letterSpacing: '0.1em',
            textTransform: 'uppercase',
            color: 'var(--sage-dark)'
          }}>
            {week.trimester === 1 ? 'First Trimester' : week.trimester === 2 ? 'Second Trimester' :
'Third Trimester'}
          </span>
        </div>

        <h1 style={{ marginBottom: 'var(--space-md)' }}>
          Week {week.weekNumber}: {week.title.split(':')[1]?.trim() || week.title}
        </h1>

        {/* Quick Stats Card */}
        <div style={{
          background: 'var(--sand)',
          borderRadius: 'var(--radius-md)',
          padding: 'var(--space-lg)',
          border: '1px solid var(--border-hairline)',
          display: 'grid',
          gridTemplateColumns: 'repeat(auto-fit, minmax(140px, 1fr))',
          gap: 'var(--space-md)',
          marginTop: 'var(--space-lg)'
        }}>
          {week.babyDevelopment.size.comparison !== 'Not yet conceived' && (
            <>
              <div>
                <p className="meta-label">Baby's Size</p>
                <p style={{
                  fontFamily: "'Playfair Display', serif",
                  fontSize: '1.25rem',
```

```
            fontWeight: 700,
            color: 'var(--charcoal)',
            margin: 0
          }}>
            {week.babyDevelopment.size.comparison}
          </p>
        </div>
        <div>
          <p className="meta-label">Length</p>
          <p style={{
            fontFamily: "'Playfair Display', serif",
            fontSize: '1.25rem',
            fontWeight: 700,
            color: 'var(--charcoal)',
            margin: 0
          }}>
            {week.babyDevelopment.size.lengthCm} cm
          </p>
        </div>
        <div>
          <p className="meta-label">Weight</p>
          <p style={{
            fontFamily: "'Playfair Display', serif",
            fontSize: '1.25rem',
            fontWeight: 700,
            color: 'var(--charcoal)',
            margin: 0
          }}>
            {week.babyDevelopment.size.weightGrams}g
          </p>
        </div>
      </>
    )}
  </div>
</header>

{/* Baby Development Section */}
<section style={{ marginBottom: 'var(--space-2xl)' }}>
  <h2>Your Baby's Development</h2>
  <p>{week.babyDevelopment.summary}</p>

  <h3 style={{ fontSize: '1.125rem', marginTop: 'var(--space-lg)' }}>Key Milestones This Week</h3>
  <ul style={{ paddingLeft: 'var(--space-xl)', color: 'var(--text-secondary)' }}>
```

```
        {week.babyDevelopment.keyMilestones.map((milestone, idx) => (
          <li key={idx} style={{ marginBottom: 'var(--space-sm)' }}>{milestone}</li>
        ))}
      </ul>
    </section>

    {/* Maternal Changes Section */}
    <section style={{ marginBottom: 'var(--space-2xl)' }}>
      <h2>Your Body This Week</h2>
      <p>{week.maternalChanges.bodyChanges}</p>

      <h3 style={{ fontSize: '1.125rem', marginTop: 'var(--space-lg)' }}>Common Symptoms</h3>
      <ul style={{ paddingLeft: 'var(--space-xl)', color: 'var(--text-secondary)' }}>
        {week.maternalChanges.symptoms.map((symptom, idx) => (
          <li key={idx} style={{ marginBottom: 'var(--space-sm)' }}>{symptom}</li>
        ))}
      </ul>
    </section>

    {/* What to Do Section */}
      <section style={{ marginBottom: 'var(--space-2xl)' }}>
      <h2>What to Do This Week</h2>

      {week.whatToDo.appointments.length > 0 && (
        <>
          <h3 style={{ fontSize: '1.125rem', marginTop: 'var(--space-lg)' }}>Prenatal
Appointments</h3>
          <ul style={{ paddingLeft: 'var(--space-xl)', color: 'var(--text-secondary)' }}>
            {week.whatToDo.appointments.map((appt, idx) => (
              <li key={idx} style={{ marginBottom: 'var(--space-sm)' }}>{appt}</li>
            ))}
          </ul>
        </>
      )}

      <h3 style={{ fontSize: '1.125rem', marginTop: 'var(--space-lg)' }}>Preparation &
Self-Care</h3>
      <ul style={{ paddingLeft: 'var(--space-xl)', color: 'var(--text-secondary)' }}>
        {week.whatToDo.preparation.map((item, idx) => (
          <li key={idx} style={{ marginBottom: 'var(--space-sm)' }}>{item}</li>
        ))}
      </ul>
    </section>
```

```
      {/* Medical Guidance Section */}
      <section style={{
        background: 'var(--sage-softest)',
        borderRadius: 'var(--radius-md)',
        padding: 'var(--space-xl)',
        border: '1px solid var(--sage-light)',
        marginBottom: 'var(--space-2xl)'
      }}>
        <h2 style={{ marginTop: 0 }}>When to Call Your Doctor</h2>
        <p style={{ marginBottom: 'var(--space-md)', color: 'var(--charcoal)' }}>
          Contact your healthcare provider immediately if you experience:
        </p>
        <ul style={{ paddingLeft: 'var(--space-xl)', color: 'var(--charcoal)', marginBottom: 0 }}>
          {week.medicalGuidance.whenToCallDoctor.map((warning, idx) => (
            <li key={idx} style={{ marginBottom: 'var(--space-sm)' }}>{warning}</li>
          ))}
        </ul>

        {week.medicalGuidance.screeningTests && week.medicalGuidance.screeningTests.length
> 0 && (
          <>
            <h3 style={{ fontSize: '1.125rem', marginTop: 'var(--space-lg)' }}>Screening Tests This
Week</h3>
            <ul style={{ paddingLeft: 'var(--space-xl)', color: 'var(--charcoal)', marginBottom: 0 }}>
              {week.medicalGuidance.screeningTests.map((test, idx) => (
                <li key={idx} style={{ marginBottom: 'var(--space-sm)' }}>{test}</li>
              ))}
            </ul>
          </>
        )}
      </section>

      {/* Fun Fact */}
      <div style={{
        background: 'var(--off-white)',
        borderLeft: '4px solid var(--sage-primary)',
        padding: 'var(--space-lg)',
        borderRadius: '0 var(--radius-sm) var(--radius-sm) 0',
        marginBottom: 'var(--space-2xl)'
      }}>
        <p style={{
          fontSize: '0.75rem',
          fontWeight: 600,
          letterSpacing: '0.1em',
```

```
    textTransform: 'uppercase',
    color: 'var(--sage-primary)',
    marginBottom: 'var(--space-xs)'
  }}>
    Did You Know?
  </p>
  <p style={{ margin: 0, fontStyle: 'italic', color: 'var(--charcoal)' }}>
    {week.funFact}
  </p>
</div>

{/* Week Navigation */}
<nav
  aria-label="Adjacent weeks"
  style={{
    display: 'flex',
    justifyContent: 'space-between',
    gap: 'var(--space-md)',
    marginBottom: 'var(--space-2xl)'
  }}
>
  {prevWeek ? (
    <Link
      to={`/pregnancy-week-by-week/week-${prevWeek}`}
      style={{
        flex: 1,
        padding: 'var(--space-lg)',
        background: 'var(--off-white)',
        border: '1px solid var(--border-hairline)',
        borderRadius: 'var(--radius-md)',
        display: 'flex',
        alignItems: 'center',
        gap: 'var(--space-sm)',
        transition: 'all var(--transition-base)'
      }}
    >
      <span style={{ fontSize: '1.5rem' }}>←</span>
      <div>
        <div style={{ fontSize: '0.75rem', color: 'var(--text-tertiary)' }}>Previous</div>
        <div style={{ fontWeight: 600, color: 'var(--charcoal)' }}>Week {prevWeek}</div>
      </div>
    </Link>
  ) : <div style={{ flex: 1 }} />}
```

```
  {nextWeek ? (
    <Link
      to={`/pregnancy-week-by-week/week-${nextWeek}`}
      style={{
        flex: 1,
        padding: 'var(--space-lg)',
        background: 'var(--off-white)',
        border: '1px solid var(--border-hairline)',
        borderRadius: 'var(--radius-md)',
        display: 'flex',
        alignItems: 'center',
        justifyContent: 'flex-end',
        gap: 'var(--space-sm)',
        transition: 'all var(--transition-base)'
      }}
    >
      <div style={{ textAlign: 'right' }}>
        <div style={{ fontSize: '0.75rem', color: 'var(--text-tertiary)' }}>Next</div>
        <div style={{ fontWeight: 600, color: 'var(--charcoal)' }}>Week {nextWeek}</div>
      </div>
      <span style={{ fontSize: '1.5rem' }}>→</span>
    </Link>
  ) : <div style={{ flex: 1 }} />}
</nav>

{/* CTA to Calculator */}
<section className="cta-inline">
  <h3>Calculate Your Current Week</h3>
  <p>Not sure which week you're in? Use our due date calculator to find out.</p>
  <Link
    to="/pregnancy-due-date-calculator"
    style={{
      display: 'inline-block',
      padding: '0.875rem 2rem',
      background: 'var(--sage-primary)',
      color: '#fff',
      borderRadius: 'var(--radius-pill)',
      fontWeight: 500,
      textDecoration: 'none',
      marginTop: 'var(--space-sm)'
    }}
  >
    Calculate Due Date →
  </Link>
```

```
      </section>

      <p className="disclaimer">
        This week-by-week guide is for informational purposes only and does not replace
professional medical advice. Always consult your healthcare provider about your specific
pregnancy.
      </p>
    </main>
  );
};
```

Phase 4: Integration & Routing

1. Update src/App.tsx

Add routes following the existing pattern:

```
// Add these imports
import { PregnancyWeekByWeekPage } from './pages/PregnancyWeekByWeek';
import { PregnancyWeekDetailPage } from './pages/PregnancyWeekDetail';

// Add these routes in the <Routes> component
<Route path="/pregnancy-week-by-week" element={<PregnancyWeekByWeekPage />} />
<Route path="/pregnancy-week-by-week/week-:weekNumber"
element={<PregnancyWeekDetailPage />} />
```

2. Update src/components/Footer.tsx

Add link to week-by-week guide in the Resources section:

```
<div>
  <h5>Resources</h5>
  <div className="site-footer__links">
    <Link to="/pregnancy-due-date-calculator">Pregnancy Due Date Calculator</Link>
    <Link to="/ovulation-calculator">Ovulation Calculator</Link>
    <Link to="/pregnancy-weight-gain-calculator">Pregnancy Weight Gain Calculator</Link>
    <Link to="/pregnancy-week-by-week">Week-by-Week Pregnancy Guide</Link> {/* NEW */}
    <Link to="/blog">Blog</Link>
  </div>
</div>
```

3. Update src/pages/Calculator.tsx

Add contextual link after results to guide users to their current week:

```
// Inside the CalculatorCard component, after results are displayed:

{dueDate && derived && (
  <>
    <ResultCard
      dueDate={dueDate}
      gestationalWeeks={derived.gestationalWeeks}
```

```
      gestationalDays={derived.gestationalDays}
      trimester={derived.trimester}
      conceptionDate={derived.conceptionDate}
    />

    {/* NEW: Week-by-week link */}
    <div style={{
      textAlign: 'center',
      margin: 'var(--space-lg) 0',
      padding: 'var(--space-md)',
      background: 'var(--sage-softest)',
      borderRadius: 'var(--radius-sm)',
      border: '1px solid var(--sage-light)'
    }}>
      <p style={{ marginBottom: 'var(--space-sm)', color: 'var(--charcoal)' }}>
        You're currently in <strong>Week {derived.gestationalWeeks}</strong>
      </p>
      <Link
        to={`/pregnancy-week-by-week/week-${derived.gestationalWeeks}`}
        style={{
          color: 'var(--sage-primary)',
          fontWeight: 600,
          textDecoration: 'underline'
        }}
      >
        View Week {derived.gestationalWeeks} Guide →
      </Link>
    </div>

    <div style={{ textAlign: 'center', margin: 'var(--space-xl) 0' }}>
      <button onClick={resetCalculator} className="primary">Calculate New Date</button>
    </div>
    {/* Continue with existing content... */}
  </>
)}
```

4. Update src/data/tools.ts

Add the week-by-week guide to the tools registry:

```
const tools: Tool[] = [
  // ... existing tools
  {
    id: 'week-by-week',
    title: 'Week-by-Week Pregnancy Guide',
    slug: 'pregnancy-week-by-week',
```

```
    description: 'Track fetal development, symptoms, and medical guidance for all 40 weeks of pregnancy.',
    path: '/pregnancy-week-by-week'
  }
];
```

5. Update public/sitemap.xml

Add entries for the hub page and all 40 week pages:

```
<!-- Add after existing URLs -->
<url>
  <loc>https://sagenest.pages.dev/pregnancy-week-by-week</loc>
  <lastmod>2026-02-22</lastmod>
  <priority>0.9</priority>
</url>
<!-- Weeks 1-40 -->
<url>
  <loc>https://sagenest.pages.dev/pregnancy-week-by-week/week-1</loc>
  <lastmod>2026-02-22</lastmod>
  <priority>0.8</priority>
</url>
<!-- ... repeat for weeks 2-40 with same priority 0.8 -->
<url>
  <loc>https://sagenest.pages.dev/pregnancy-week-by-week/week-40</loc>
  <lastmod>2026-02-22</lastmod>
  <priority>0.8</priority>
</url>
```

6. Update README.md

Add to the Core Features section:

- Week-by-Week Pregnancy Guide with 40 detailed weekly guides covering:
  - Fetal development milestones and size comparisons
  - Maternal body changes and common symptoms
  - Prenatal appointments and screening tests
  - Preparation tips and medical guidance
  - Red flag symptoms requiring immediate attention

Content Writing Guidelines

For all 40 weeks, follow these content principles:

Tone & Style

Calm and reassuring: Pregnancy can be anxiety-inducing; content should be factual but comforting

Medical accuracy: All development milestones must align with ACOG guidelines

Scannable: Use short paragraphs, bullet points, and clear headings

Actionable: Each week should have 2-3 concrete things the user can do

Size Comparisons (Baby Development)
Use familiar fruits/objects that accurately represent size:
Weeks 1-4: Pre-conception → poppy seed
Weeks 5-8: Sesame seed → raspberry
Weeks 9-13: Cherry → lemon
Weeks 14-18: Avocado → bell pepper
Weeks 19-24: Mango → corn
Weeks 25-30: Cauliflower → cabbage
Weeks 31-36: Pineapple → honeydew melon
Weeks 37-40: Watermelon
Medical Milestones to Include
Week 10-13: First trimester screening, NT scan availability
Week 15-20: Maternal serum screening, quickening may begin
Week 20-22: Anatomy scan (mention this is when many find out sex)
Week 24-28: Glucose challenge test (gestational diabetes screening)
Week 28: Begin bi-weekly appointments
Week 32: May begin weekly appointments depending on provider
Week 35-37: Group B strep test
Week 36+: Weekly appointments standard
Week 37: Full term status
Week 39: Brain development complete
Week 40: Estimated due date
Week 41+: Post-dates monitoring
Red Flags (When to Call Doctor)
Always include for each week, tailored to that week's risks:
Heavy bleeding (soaking pad in <1 hour)
Severe abdominal pain
Fever >100.4°F (38°C)
Sudden severe swelling (face/hands)
Severe headache with vision changes
Decreased fetal movement (after week 20)
Fluid leaking (any week)
Contractions before 37 weeks
SEO Requirements
Meta Titles (follow exact pattern)
Hub page: "Week-by-Week Pregnancy Guide — SageNest Health – Smart Tools for Women's Wellness"
Week pages: "Week [N] of Pregnancy: [Short Title] — SageNest Health – Smart Tools for Women's Wellness"
Meta Descriptions (150-160 characters)
Hub: "Track your pregnancy week by week with detailed fetal development, symptoms, medical guidance, and preparation tips for all 40 weeks."
Week: "Week [N] pregnancy guide: [Baby size], key milestones, symptoms, prenatal appointments, and medical guidance. Evidence-based pregnancy tracking."

JSON-LD Structured Data

Every week page needs:

```
{
  "@context": "https://schema.org",
  "@type": "Article",
  "headline": "Week [N] of Pregnancy: [Title]",
  "description": "[Baby development summary]",
  "datePublished": "2026-02-22",
  "dateModified": "2026-02-22",
  "mainEntityOfPage": "https://sagenest.pages.dev/pregnancy-week-by-week/week-[N]",
  "publisher": {
    "@type": "Organization",
    "name": "SageNest",
    "url": "https://sagenest.pages.dev"
  }
}
```

Internal Linking Strategy

Hub page links to: all 40 weeks, due date calculator

Each week page links to: hub page, previous week, next week, due date calculator

Due date calculator links to: hub page + user's current week

Footer links to: hub page

Accessibility Requirements (WCAG 2.1 AA)

Navigation

Use <nav> with aria-label="Breadcrumb" for week pages

Week grid should use semantic links with clear link text

Previous/Next navigation should use aria-label="Adjacent weeks"

Content Structure

Heading hierarchy: h1 (page title) → h2 (sections) → h3 (subsections)

Lists use proper <ul> and <li> markup

No heading levels skipped

Interactive Elements

All buttons have clear, descriptive text

Focus states visible (2px outline, 3px offset)

Tab order follows visual order

Color is not the only means of conveying information

Testing Checklist

[ ] All images have alt text (if images added later)

[ ] Keyboard navigation works (Tab, Enter, Arrow keys)

[ ] Screen reader announces all content properly

[ ] Color contrast meets 4.5:1 minimum for normal text

[ ] Focus indicators visible on all interactive elements

Testing & QA Checklist

Before considering this feature complete:

Functionality Tests

[ ] Hub page renders all 40 weeks correctly
[ ] Trimester filter works (all/first/second/third)
[ ] "Find Your Week" input accepts 1-40 and navigates correctly
[ ] "Find Your Week" input rejects invalid input (0, 41, letters)
[ ] Each week page displays correct data
[ ] Previous/Next navigation works on all week pages
[ ] Week 1 has no "Previous" button
[ ] Week 40 has no "Next" button
[ ] Links from calculator to current week work
[ ] Breadcrumb navigation works
[ ] All internal links resolve correctly

Visual Tests

[ ] Hub page matches design system (colors, spacing, typography)
[ ] Week pages match design system
[ ] Week grid displays properly at all viewport sizes
[ ] Mobile view: week cards stack properly
[ ] Mobile view: tabs wrap or scroll horizontally
[ ] Quick stats card responsive on mobile
[ ] Medical guidance callout box stands out visually

SEO Tests

[ ] All pages have unique meta titles
[ ] All pages have unique meta descriptions (150-160 chars)
[ ] Canonical URLs set correctly
[ ] JSON-LD validates (use Google Rich Results Test)
[ ] Sitemap includes all 41 pages (hub + 40 weeks)
[ ] All internal links use correct paths

Accessibility Tests

[ ] All pages pass automated accessibility checker
[ ] Keyboard navigation works throughout
[ ] Screen reader announces all content
[ ] Focus indicators visible
[ ] Heading hierarchy correct (no skipped levels)
[ ] ARIA labels used where appropriate

Performance Tests

[ ] Hub page loads in <3 seconds
[ ] Week pages load in <2 seconds
[ ] No console errors
[ ] Images optimized (if added)

Content Tests

[ ] All 40 weeks have complete data
[ ] Size comparisons accurate

[ ] Medical milestones align with ACOG guidelines
[ ] Red flags appropriate for each week
[ ] No typos or grammatical errors
[ ] Tone is consistent and reassuring
Build & Deployment
Local Testing

```
# Install dependencies (if needed)
npm install

# Run dev server
npm run dev

# Test routes manually:
# http://localhost:5173/pregnancy-week-by-week
# http://localhost:5173/pregnancy-week-by-week/week-1
# http://localhost:5173/pregnancy-week-by-week/week-20
# http://localhost:5173/pregnancy-week-by-week/week-40

# Run TypeScript check
npm run build

# Run tests
npm run test
```

Deployment Checklist
[ ] All TypeScript compiles without errors
[ ] All tests pass
[ ] npm run check:package passes
[ ] npm run check:conflicts passes
[ ] Git commit message follows convention: feat(week-by-week): add pregnancy week-by-week guide with 40 detailed pages
[ ] Push to GitHub triggers Cloudflare build
[ ] Verify deployment on Cloudflare Pages
[ ] Test all routes in production
[ ] Submit sitemap to Google Search Console
Success Criteria
This feature is considered complete when:
All routes work: Hub page + all 40 individual week pages load without errors
Navigation functions: All internal links, previous/next buttons, breadcrumbs work
Calculator integration: Due date calculator shows link to user's current week
Design consistency: Matches existing SageNest design system exactly
SEO optimized: All meta tags, JSON-LD, sitemap entries present
Accessibility compliant: Passes WCAG 2.1 AA automated tests

Mobile responsive: Works perfectly on 320px to 1920px viewports
Content complete: All 40 weeks have medically accurate, well-written content
No regressions: Existing calculator, blog, tools pages still work
Build succeeds: npm run build completes without errors

Final Notes
Follow SECURITY.md Anti-Mess Protocol: Make surgical changes, touch only what's needed
Match existing patterns: Look at Calculator.tsx, OvulationCalculator.tsx for reference
Use design tokens: Always use CSS variables from global.css
Test incrementally: Test each phase before moving to the next
Ask if unsure: Don't assume; surface confusion explicitly
Remember: This is the single highest-ROI feature for SageNest. Take time to do it right. Quality over speed.